ORBIT PULSE

Satellite Conjunction Risk Intelligence
& Decision Support

FINAL PROTOTYPE MASTER BLUEPRINT

Concept-led by the research blueprint • Implementation-led by the SW-5 prototype specification

A build-ready specification for a polished, auditable hackathon prototype with live-updating risk graphs, event-level forecasting, evidence intelligence and transparent decision support.

Single source of truth for a three-person beginner team. This document defines what to build, how it should behave, how to evaluate it, what the interface should show, and what must not be claimed.

Important: Charts can update as new CDMs arrive or a replay advances. A genuine live operational CDM feed is not assumed; it must be authorised and technically available. Without it, use historical replay or manual refresh and label the mode clearly.

Version 1.0 • 7 November 2026 • Student research prototype — not for operational spacecraft manoeuvre decisions

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1. Executive direction and combined concept

The two source documents are merged with a deliberate division of responsibility: the ORBIT PULSE research blueprint supplies the product concept, differentiation and extended lifecycle vision; the SW-5 final prototype document supplies the implementation plan, data pipeline, model workflow, dashboard structure, tests and acceptance criteria. This is one coherent build specification, not two competing plans.

Project definition

ORBIT PULSE is a time-aware satellite conjunction risk intelligence platform. It reconstructs each conjunction event from its Conjunction Data Message (CDM) history, builds a strictly leakage-safe snapshot from information available no later than 48 hours before Time of Closest Approach (TCA), forecasts the dataset-defined final reported collision probability (Pc), and presents an auditable review recommendation. Its Evidence Intelligence Layer makes data freshness, completeness and forecast stability visible.

Core pitch

> “Space traffic management is not only about estimating close approaches; it is about making decisions from incomplete and changing information. ORBIT PULSE forecasts final reported conjunction risk from early CDM history, shows how fresh and complete the evidence is, exposes prediction stability, and connects the forecast to a transparent review workflow.”

Credible student contribution

- Strict early-CDM forecasting at the TCA–48-hour boundary.
- Auditable comparison against a simple last-observation baseline.
- Evidence Intelligence: freshness, completeness and forecast sensitivity displayed separately from predicted Pc.
- Decision policy exposes missed-dangerous-event vs unnecessary-review trade-offs.
- Interactive dashboard with live-updating event graphs, traceable cutoffs and model audit information.
- Optional scenario sandbox for candidate manoeuvre screening and post-action verification.

2. Problem definition and objectives

A conjunction event occurs when two tracked space objects are predicted to pass close to one another. CDMs provide successive estimates of the encounter. As tracking and orbit solutions are updated, miss distance and Pc can change. The challenge is to estimate eventual dataset-reported risk using only information available at least two days before TCA.

Primary prediction task

For each eligible event, predict the **final reported CDM Pc** (or exact competition-defined final risk target) from an event snapshot containing only messages whose approved creation/availability timestamp is at or before TCA–48 hours. The final target is a training/evaluation label only and must never enter the feature table.

Target distinction

A final CDM Pc is a final reported risk estimate, not evidence that a collision actually occurred. Unless verified collision outcomes exist, do not claim to predict actual collisions. If organisers define a binary dangerous-event class instead of numerical Pc, document that definition and make it the primary task. Do not silently substitute targets.

Objectives

1. Load approved historical CDM data and validate schema, timestamps, units and event identifiers.
2. Reconstruct chronological histories for each event without mixing events.
3. Enforce TCA–48h and create one event-level snapshot for each eligible event.
4. Train and compare a simple baseline and suitable ML model(s).
5. Evaluate on chronological, event-separated data and report rare high-risk errors.
6. Display predicted risk, evidence indicators, recommendation logic and risk history.
7. Provide reproducible metadata and automated cutoff/leakage tests.

Success is not the most complicated model. Success is a valid target, correct temporal logic, honest evaluation, and a reviewer tracing a prediction back to its exact event snapshot.

3. Scope, assumptions and differentiation

In scope

- CSV ingestion first; add XML/KVN only after confirming a real sample schema.
- Validation, event grouping, duplicate handling, timestamp ordering and 48-hour snapshots.
- Feature engineering, baseline, one or two candidate models, threshold selection and evaluation.
- Streamlit dashboard: overview, explorer, event detail, explanation, performance and audit.
- Evidence Intelligence and live-updating charts driven by incoming data or replay.
- Optional scenario sandbox only if sufficient state/orbit data and a suitable propagator exist.

Out of scope

- Autonomous spacecraft control, burn-command generation or operational approval.
- Replacing NASA/ESA or commercial conjunction assessment systems.
- Claims of first-ever collision prediction or industry-first innovation.
- Defensible Pc from miss distance alone without covariance, encounter-plane geometry, hard-body radius and validated method.
- TLE/OMM as a substitute for historical CDM event sequences.
- Claims that very small debris is individually tracked.

Honest differentiation

Agencies and operators already perform space surveillance, conjunction assessment, orbit uncertainty analysis, manoeuvre planning and post-manoeuvre checks. ORBIT PULSE's credible contribution is a transparent student prototype combining early-CDM forecasting, strict information cutoff, evidence freshness/completeness/stability, baseline comparison and a traceable review workflow. Novelty is the integration and evaluation, not claiming established aerospace methods as inventions.

4. Product experience and visual design specification

Visual target: a professional mission-control analytics console. Use a dark navy/charcoal canvas, restrained cyan/blue accents, clear typography, generous spacing, strong chart hierarchy and minimal decoration. Polish must not obscure data provenance or limitations.

Element	Specification
Style	Mission operations / aerospace analytics; dark theme default with accessible contrast.
Palette	Deep navy background, slate panels, cyan/blue data accents; green/amber/red only for meaningful states.
Typography	Clean sans-serif; large KPI numbers, compact labels, readable axes.
Layout	Left navigation; top status bar; main content grid; consistent cards and chart panels.
Motion	Subtle loading/update transitions only; no animation implying unsupported precision or urgency.
Responsive	Desktop-first for judging; usable on laptop and narrower browser.
Accessibility	Color is never the only cue; include text labels, legends, units and time zone.

Required screens

Screen	Must show	Visual treatment
Mission Overview	Dataset/version, event counts, eligible/ineligible, review count, data mode, model version and metrics.	KPI cards; risk distribution; update timeline; model status.
Live Risk Monitor	Incoming/replayed updates, event risk trends, message age, flagged events, feed status.	Auto-refreshing trends; event list; last update; LIVE/REPLAY/MANUAL badge.
Event Explorer	Search/filter event ID, TCA, object IDs if available, Pc/risk band and recommendation.	Sortable table; risk labels; eligibility and data-quality indicators.
Event Detail	Metadata, chronological history, cutoff marker, predicted Pc, allowed count and recommendation.	Time-series chart with TCA–48h boundary; eligible/excluded messages distinguished.
Evidence & Explanation	Freshness, completeness, stability, feature explanation and threshold logic.	Three independent indicator panels; sensitivity plot.
Model Performance	Baseline vs ML, chronological split, confusion matrix, PR curve, error distribution and ablation.	Comparison cards; PR curve; confusion matrix; calibration only if justified.
Audit & Data Quality	Dataset/schema/model versions, cutoff, exclusions, duplicates, missingness and config.	Audit timeline; validation summary; report export.

Visual priority order

1. Is the data current, and is this LIVE, REPLAY or MANUAL?
2. Which events need review, and why?
3. What information was available at TCA–48h?
4. What did the model predict, how stable is it, and what rule produced the recommendation?
5. How well does the system perform on held-out chronological data?

Every graph needs a question it answers, title, units, time zone, legend, source context and last-updated time.

5. Live-updating graphs and data modes

The prototype must include live-updating visualisations. Charts refresh when a new CDM record arrives, a user refreshes a dataset, or a replay stream advances. This does **not** imply access to a guaranteed public real-time operational CDM feed. Access, permissions, rate limits and availability must be confirmed independently.

Mode	Behaviour	Required label
LIVE FEED	Approved source only; poll/subscribe at documented interval; validate each record; update event histories and charts.	LIVE • source • last received timestamp
REPLAY	Play historical CDM sequence in timestamp order at selectable speed; reliable demo mode.	REPLAY • historical data • simulated clock
MANUAL / BATCH	Upload/select permitted CSV; rebuild snapshots and refresh charts on demand.	MANUAL DATA • dataset version • load time

Required interactive graphs

Graph	Purpose	Interaction
Event Pc history	Reported Pc over time; logarithmic y-axis often appropriate.	Hover; zoom; range selector; cutoff marker; eligible vs later messages.
Risk monitor timeline	Events in low/review/high bands over time or update batch.	Toggle band; select interval; click to open event.
Event update stream	CDM arrivals by time/event/source; not a risk metric.	Pause/resume replay; playback speed; event filter.
Forecast sensitivity	Prediction range after eligible-message perturbations.	Base forecast + range; label model sensitivity, not orbit uncertainty.
PR curve	Precision-recall trade-off if valid binary target exists.	Hover; mark validation-selected threshold.
Confusion matrix	TP/FP/TN/FN at selected threshold.	Show counts/class definitions; inspect false negatives.
Feature contribution	How model output relates to important inputs.	Name attribution method; importance is not causation.

Live update rules

- Persistent LIVE / REPLAY / MANUAL badge; never disguise replay as live.
- Show last received timestamp and feed age. No update → STALE / NO NEW DATA.
- Only validated records update charts; malformed messages are quarantined and counted.
- Do not retrain on every incoming message. Ingestion updates views/predictions; retraining is controlled.
- Every prediction stores event ID, input cutoff, dataset snapshot/version, model version and prediction timestamp.
- A later CDM can appear in the event timeline but must not change the locked TCA–48h feature snapshot.

Recommended: Streamlit + Plotly. Use a controlled refresh/rerun pattern for periodic updates and sorted replay for demos. Add WebSocket/FastAPI only if a real integration requirement justifies it.

6. System architecture

Layer	Responsibility	Input → Output
CDM Data Intelligence	Ingest, validate, normalise, group, sort, deduplicate, audit.	Raw CDM/CSV → validated event histories
48-hour Snapshot Builder	Apply official timestamp rule and lock eligible records.	History + TCA → leakage-safe event snapshot
Risk Forecasting Engine	Features; baseline and ML; final-Pc estimate/class score.	Snapshot → forecast + metadata
Evidence Intelligence	Freshness, completeness, forecast sensitivity.	Snapshot + perturbation runs → separate indicators
Decision Support	Documented threshold/cost policy; advisory label.	Forecast + evidence → review status
Scenario Sandbox (optional)	No-action vs candidate Δv; secondary screening if supported.	State/ephemeris → scenario comparison
Post-action Verification (optional)	Re-screen using updated actual/simulated state.	Updated state → before/after assessment
Dashboard & Audit	Live/replay/manual status, charts, event detail, performance, provenance.	All module outputs → reviewer interface

End-to-end data flow

1. Load approved data and assign dataset version.
2. Validate columns, parse timestamps, normalise units and log issues.
3. Group by stable event ID and sort by approved availability timestamp.
4. Compute TCA–48h boundary and select eligible messages only.
5. Build one event feature row; keep final target isolated as label.
6. Train/evaluate with chronological event splits; choose model/threshold on validation.
7. Save model, preprocessing, configuration, metrics and metadata.
8. At inference, snapshot → predict → evidence indicators → decision policy.
9. Render dashboard and append traceable audit record.

7. CDM data model and 48-hour cutoff

Data feasibility is the first engineering gate. Obtain a permitted sample and confirm multiple messages per event, stable event identifier, TCA, approved creation/availability timestamp and required final target. If these are absent, document the limitation; do not fabricate model results.

Logical field	Purpose	Requirement
event_id	Links all messages for one conjunction event.	Required
message_time	Timestamp determining when a message was available.	Required; official semantics must be confirmed
TCA	Time of closest approach.	Required
final_target	Final reported Pc or official risk label.	Required for supervised evaluation; label only
Pc / miss distance / relative velocity	Potential predictors when present in eligible messages.	Optional; document units
Covariance / uncertainty	Potential encounter uncertainty descriptors.	Optional; preserve frames and conventions
Object IDs / metadata	Event exploration and grouping context.	Optional
Source/schema version	Traceability and parsing.	Recommended

Cutoff rule

Default interpretation, subject to official rules: **eligible message_time** ≤ **TCA** − **48 hours**. A message exactly at the boundary is included if the official rule is inclusive. Any later message is excluded from model inputs. Confirm whether the official field is creation time, receipt time or another availability timestamp.

Pseudo-logic

For each event: sort messages by approved timestamp; compute cutoff = TCA−48h; eligible = messages with timestamp ≤ cutoff; if empty, mark event ineligible; otherwise compute features only from eligible messages. Never substitute a later message for an event with no eligible observation.

Validation and governance

- Validate timestamps, time zones, TCA consistency, plausible ranges and units.
- Count duplicates; use documented deduplication keys and retain audit counts.
- Detect inconsistent event IDs/TCA/object pairing; quarantine unresolved records.
- Track missing fields, invalid units, unknown coordinate frames and schema changes.
- Keep raw files immutable; cleaned data in separate processed storage.
- Keep credentials outside source control; do not publish restricted data.
- Never mix different event IDs, even if object identifiers overlap.

8. Features and machine learning

Create one feature row per eligible event. Every feature must be computable from eligible records only. Actual feature availability depends on the dataset; do not assume a standard CDM field is present.

Family	Candidate features	Notes
Pc history	First/last/min/max/mean/std/count; log10(Pc+ε); trend/slope.	Only if Pc exists in eligible messages; handle zeros/missingness carefully.
Geometry	Miss distance, relative speed, encounter time.	Use documented units and frames.
Uncertainty	Covariance summaries, uncertainty magnitudes, encounter-plane descriptors.	Do not invent fields or derive Pc from geometry alone.
Temporal	Time to TCA at cutoff, latest eligible message age, update intervals/count.	Use official timestamp.
Data quality	Missingness, valid-message count, schema completeness.	No post-cutoff quality information.
Context	Object class/metadata; optional time-aligned space weather.	Add only if justified.

Target handling

- Regression: predict final reported Pc; log10(Pc+ε) may suit its wide dynamic range. Document ε, zero handling and inverse transform.
- Classification: use the official dangerous-event class definition exactly.
- Never mix regression target and classification label into one ambiguous metric.

Models in order

Order	Model	Purpose
1	Last eligible Pc baseline (or official simple baseline).	Essential benchmark.
2	Linear/Ridge regression on log-Pc features, if regression target.	Simple trainable baseline.
3	Random Forest or HistGradientBoosting / Gradient Boosting.	Nonlinear tabular candidate.
4 optional	XGBoost / LightGBM / CatBoost.	Only if time, install and validation benefit justify it.

Imbalance and interpretability

For a valid rare dangerous-event class, consider class weights inside the training pipeline. Never oversample before splitting. Select thresholds on validation data based on missed-danger vs unnecessary-review cost; never tune on test data. Explain predictions with concise feature contributions or transparent rules. Feature importance is not causation; use permutation/model-native importance if SHAP is too costly.

9. Evaluation and research hypotheses

Use event-level splits, not random message-level splits. No event may appear in more than one split. Test must represent a later time period than training. Fit preprocessing, imputation, scaling, feature selection and threshold selection using training/validation only.

Recommended split

Sort events by a defensible event timestamp (e.g. TCA or event creation date as appropriate), then divide chronologically into train, validation and held-out test periods. Record exact date ranges, counts and class counts. Small datasets require explicit uncertainty and cautious generalisation claims.

Task	Primary metrics	Additional reporting
Pc regression	MAE and RMSE in log10(Pc) space.	Median absolute error; risk-band error; baseline-relative improvement; error distribution.
Dangerous classification	Recall, precision, false negatives, false positives, PR-AUC.	Confusion matrix; ROC-AUC supplementary; threshold and class definition.
Calibration	Brier score/reliability curve only for valid class probabilities.	Calibration method and validation protocol; never label uncalibrated score as probability.
Review burden	Number escalated and unnecessary escalations under stated label definition.	Always report alongside missed dangerous events.

Mandatory ablation

Experiment	Feature set	Question
A	Last eligible Pc only	How strong is baseline?
B	Pc history summaries	Does history add value?
C	History + available geometry/uncertainty	Does encounter context help?
D	History + geometry + evidence features	Do evidence features add value?

Research hypotheses

- H1: CDM-history features available at TCA–48h predict final reported Pc better than last-observation baseline on chronological held-out data.
- H2: freshness/stability indicators identify materially changing forecasts; test whether they add value beyond risk forecast.
- H3: transparent review policy exposes missed-high-risk vs unnecessary-escalation trade-off.

- H4: candidate manoeuvre screening can reveal secondary conjunctions when adequate state data and propagator exist.

Reporting rules

- Report dataset size, time span, eligible count and target distribution.
- Report exclusions and reasons.
- Include baseline even if ML is worse.
- Inspect false negatives and positives.
- Never fabricate performance; synthetic data supports UI/pipeline demos only.
- Keep model metrics separate from individual event risk.

10. Evidence Intelligence

This is the signature concept feature. Do not collapse it into an arbitrary “confidence percentage.” Display three independent dimensions.

Indicator	Definition	Display
Data Freshness	Age of latest eligible message/orbit solution.	Age in hours + fresh/ageing/stale label; configured thresholds.
Evidence Completeness	Availability of expected fields and usable history.	Completeness ratio/checklist; required vs optional fields separate.
Forecast Stability	Sensitivity to reasonable perturbations of eligible input history.	Base forecast + min/max or percentile range; stability label.

Sensitivity experiment

1. Predict from complete eligible history.
2. Repeat after removing one eligible message at a time where history permits.
3. Optionally perturb eligible numeric inputs within documented data-quality tolerances.
4. Compare predictions with base; report range/spread/relative change.
5. Label “model sensitivity to eligible input history.” It is not calibrated physical Pc, orbit covariance or a real-world uncertainty bound.

Interpretation

High forecast + stale/incomplete evidence → HIGH — REVIEW; EVIDENCE LIMITED. High forecast + fresh/stable evidence → HIGH — REVIEW; EVIDENCE STRONGER. Low forecast + unstable evidence must not look like an all-clear. Indicators guide review; they do not replace predicted Pc.

11. Decision policy

Keep prediction engine and decision policy separate. The model estimates the dataset-defined target; a separate layer applies documented threshold/cost assumptions. Select thresholds on validation data, never the final test set.

Label	Meaning	UI wording
LOW	Below documented threshold and evidence adequate for stated policy.	LOW — ROUTINE MONITORING
UNCERTAIN	Stale/incomplete evidence, unstable model, or out-of-domain event.	UNCERTAIN — FURTHER ASSESSMENT
HIGH	Forecast meets/exceeds documented review threshold.	HIGH — ESCALATE FOR MANOEUVRE REVIEW

Do not say “execute manoeuvre,” “collision certain,” or “safe to ignore.” This is advisory. Qualified operators make real operational decisions. If competition requires binary MANOEUVRE / NO MANOEUVRE, show it as competition output while retaining evidence qualifier and disclaimer.

12. Manoeuvre sandbox and post-action verification

Advanced extension only after the core predictor and dashboard work.

Scenario sandbox

- Compare no-action reference trajectory with user-supplied candidate Δv scenarios.
- With validated propagator and adequate state/ephemeris, estimate closest-approach geometry.
- Screen candidate trajectory against catalogued objects for secondary conjunctions if supported.
- Estimate Δv/fuel only when mass, propulsion parameters and assumptions are known; otherwise show Δv as input.
- Do not claim Pc from geometry alone; Pc requires covariance, encounter-plane analysis, hard-body radius and validated method.

Post-action verification

After actual/simulated manoeuvre, ingest updated orbit/state and rerun screening. Compare original geometry, identify new close approaches and state data horizon/method. A commanded Δv is not proof of achieved orbit; use updated actual or explicitly simulated state.

Optional debris-protection study

Very small debris/micrometeoroids are often not tracked individually. Their protection is an engineering/environment issue. A separate study may explain Whipple shielding (bumper, standoff gap, rear wall), protected placement and speculative reconfigurable panels. Whipple shielding is established, not new. Folding arrays/antennas are not universal protection: power, thermal, communications, attitude, reliability and coverage trade-offs must be considered. Keep separate from PS-5 scoring.

13. Data sources and technology stack

Data acquisition order

1. Ask StarForge organisers/mentor for approved dataset, sample CDM XML/KVN/CSV, event key, final target, timestamp rule, licence and demo permission.
2. Check Space-Track account, terms and actual CDM access. Registration does not guarantee historical CDMs; verify current official documentation before coding queries.
3. Use CelesTrak TLE/OMM only for limited orbit/context demos; it is not a substitute for CDM history or final Pc labels.
4. Synthetic data may test pipeline/UI only, clearly labelled; it cannot substantiate real-CDM predictive performance.

Source	Role	Caveat
Space-Track	Potential CDM/orbit data access.	Access, data classes, terms and fields must be verified.
CCSDS	CDM/ODM standards.	Standards are not historical training data.
NASA CARA / Orbital Debris	Conjunction and debris methodology context.	Research context; no implied endorsement.
ESA Space Safety / DISCOS	Debris context and object metadata.	Coverage/API access vary.
CelesTrak	TLE/OMM orbit elements.	Not CDM history.
NOAA SWPC / NASA OMNIWeb	Optional space-weather context.	Only add if aligned and justified.

Recommended stack

Area	Choice	Reason
Language	Python	Fast iteration and ML ecosystem.
Data	pandas, NumPy	Cleaning, timestamps, aggregation.
ML	scikit-learn	Pipelines, baselines, tree models, metrics.
Charts	Plotly	Interactive zoom, hover, time series.
Dashboard	Streamlit	Fastest working hackathon app.
Storage	CSV/Parquet; SQLite optional	Keep infrastructure light.
Models	joblib + JSON metadata	Persist model and provenance.
Orbital tools	Orekit advanced; Skyfield limited context	Only with validated inputs/method.
API/frontend	FastAPI/React optional	Do not build before Streamlit MVP.

Repository layout

ORBIT-PULSE/
■■■ data/raw/ processed/ demo/ (do not commit restricted data)
■■■ models/ risk_model.joblib model_metadata.json
■■■ notebooks/ exploration.ipynb
■■■ src/ data_loader.py validation.py event_builder.py snapshot_48h.py
■ feature_engineering.py train_model.py evaluate_model.py predict.py
■ evidence_intelligence.py decision_engine.py audit.py
■■■ dashboard/ app.py pages/ components/ charts.py theme.py
■■■ tests/ test_cutoff.py test_event_grouping.py test_no_leakage.py
■ test_timestamps.py test_decision_policy.py
■■■ configs/ feature_schema.yaml thresholds.yaml
■■■ reports/ data_quality.md metrics.json
■■■ requirements.txt README.md .env.example .gitignore

14. Team roles and schedule

Person	Ownership	Deliverables
1 — Data & ML	Dataset, schema, parser, event grouping, cutoff, features, baseline and model.	Data report; snapshot pipeline; model; metrics; error review.
2 — Evidence/domain	Freshness, completeness, stability; decision policy; scenario research/caveats.	Evidence module; decision rules; scenario feasibility note.
3 — Dashboard/integration	Streamlit, Plotly, live/replay/manual modes, event explorer, audit, integration.	Polished dashboard; charts; demo; README.

Coordination rule

Agree on one event-level schema and function interfaces before parallel coding. Person 1 publishes snapshot columns and prediction output contract early. Persons 2 and 3 can build against labelled fixtures until real data arrives.

Priority	Build	Exit condition
P0 feasibility	Confirm dataset, labels, timestamps, histories, licence.	Sample proves target and pre-cutoff history.
P0 pipeline	Validate, group, sort, deduplicate, cutoff snapshots.	Tests pass; one audited row per eligible event.
P0 model	Last-Pc baseline, candidate model, chronological evaluation.	Honest metrics and false-negative review.
P0 dashboard	Overview, event detail, cutoff marker, prediction, recommendation.	One event traceable end-to-end.
P1 visual/live	Plotly, LIVE/REPLAY/MANUAL, update status.	Charts update honestly.
P1 evidence	Freshness, completeness, stability and explanation.	Three separate indicators.
P1 audit	Versions, exclusions, model metadata, reproducibility.	Prediction provenance traceable.
P2 scenario	Candidate Δv, secondary screening if supported.	Validated assumptions and limitations.
P3 debris study	Optional shielding/folding-panel design study.	Separate from predictor.

3–5 day sprint

1. Day 1: obtain sample; confirm target/timestamps; parser, event grouping, cutoff tests.
2. Day 2: feature table, last-Pc baseline, one tree model, chronological evaluation and error review.
3. Day 3: Streamlit overview + event detail + Plotly history + cutoff marker + recommendation.
4. Day 4: replay/live-update simulation, evidence indicators, audit panel and visual polish.
5. Day 5: integration, acceptance tests, demo rehearsal and documentation.

Do not sacrifice cutoff correctness or evaluation to add 3D orbit animation. An accurate interactive timeline is more valuable than scientifically unsupported spectacle.

15. Tests and demo script

Test	Expected result
Cutoff boundary	Exactly TCA–48h included if inclusive; later message excluded.
No eligible message	Event marked ineligible; no later substitute.

Test	Expected result
Event grouping	Different event IDs never mix.
Timestamp order	History sorted by approved availability timestamp.
Duplicates	Consistent handling and audit count.
Leakage	No post-cutoff or target-derived predictor.
Split integrity	No event in multiple splits; test later than train.
Reproducibility	Seeds/configuration/model metadata saved.
Threshold	Selected on validation only.
Traceability	Event ID, cutoff, dataset/model version, prediction time shown.
Live/replay	Replay labelled; live source and last-received time shown.
Stale feed	No update → stale/no-update state.
Safe wording	Never an autonomous command.

Demo-ready definition

Reviewer loads/selects approved data, sees validation summary, selects event, inspects chronological history and cutoff, views prediction/evidence, understands recommendation threshold, and inspects evaluation without manually editing data.

3–5 minute demo

1. Introduce early risk estimation using only information available at least 48 hours before TCA.
2. Open overview; show dataset status, eligible/ineligible counts and data mode.
3. Open Live Risk Monitor; show graph changing as replay advances or approved data arrives.
4. Select event; show CDM history and TCA–48h cutoff; explain excluded messages.
5. Show snapshot, predicted final Pc, freshness, completeness and stability.
6. Explain review label and validation-selected threshold.
7. Show baseline vs ML, recall, false negatives, false positives and PR-AUC where applicable.
8. Close with target/data limitations and non-autonomous disclaimer.

16. Risks and final submission checklist

Risk	Mitigation
Dataset unavailable/incomplete	Verify first; ask organisers; no fabricated performance; synthetic only for labelled UI/pipeline.
No defensible target	Ask organisers; disclose any alternative label.
Few dangerous examples	Appropriate metrics, class weights, cautious validation.
True collisions rare	Describe dataset-defined Pc target; do not imply actual collision prediction.
Leakage	Strict event cutoff, chronological split, feature audit and tests.
Poor calibration	Evaluate only if meaningful; distinguish risk, uncertainty and confidence.
No live feed	Clearly labelled replay; never call replay operational live data.
Operational misuse	Advisory wording, disclaimer, human review.
Scope creep	Complete P0 before scenario, APC, React/API or advanced animation.

Final submission checklist

- Dataset source/permission/version, target, timestamps and units documented.
- Cutoff implementation and boundary tests pass.
- Grouping, duplicates, ineligible-event and leakage tests pass.
- Baseline and model evaluated on chronological event-separated test data.
- False negatives/positives reviewed; threshold selected on validation.

- Dashboard has overview, mode status, explorer, event detail, evidence, performance and audit.
- Graphs have labels, units, legends, time range and update context.
- Evidence indicators separately defined; no arbitrary confidence percentage.
- Model/data/config versions and prediction timestamps traceable.
- Limitations and operational disclaimer visible; demo runs cleanly.

Final proposed solution statement

ORBIT PULSE is a time-aware satellite conjunction risk intelligence prototype that reconstructs historical CDM sequences and creates an auditable event snapshot using only information available at or before TCA–48 hours. It forecasts the dataset-defined final reported conjunction risk, compares performance with a simple baseline using event-separated chronological evaluation, and presents a transparent review recommendation. Its Evidence Intelligence Layer separately exposes data freshness, evidence completeness and forecast stability, while an interactive dashboard provides live-updating or replay-driven risk graphs, event history, model performance and audit traceability. Optional manoeuvre screening and post-action verification are included only where adequate orbit/state data and validated methods are available. It is a research and decision-support prototype, not autonomous spacecraft control.

Appendix A. Data contract and core definitions

Field	Type / example	Meaning
event_id	string	Stable conjunction event key.
tca_utc	ISO-8601	TCA in UTC.
cutoff_utc	ISO-8601	TCA minus 48 hours.
latest_eligible_message_utc	ISO-8601	Latest admitted message.
eligible_message_count	integer	Messages used.
features	numeric/categorical	Derived only from eligible messages.
target_final_pc	numeric/null	Training label only; excluded from predictors.
eligible	boolean	At least one eligible message.
exclusion_reason	string/null	Why event is ineligible.
dataset_version	string	Source/version.
feature_schema_version	string	Feature contract version.

Core definitions

- Cutoff = TCA–48h.
- Eligible set E = messages for event e with approved message_time ≤ cutoff.
- Freshness age = prediction timestamp – latest eligible message timestamp.
- Completeness = present expected fields / documented expected field count; separate required from optional.
- Stability = prediction dispersion across documented eligible-history perturbations.
- Regression error = MAE/RMSE on log10(Pc+ε), with ε and zero handling documented.
- Decision threshold = validation-selected using documented cost assumptions.

Prediction output contract (illustrative placeholders)

```
{
"event_id": "event-key",
"prediction_target": "final_reported_pc",
"predicted_value": null,
"risk_band": "LOW | UNCERTAIN | HIGH",
"recommendation": "ROUTINE_MONITORING | FURTHER_ASSESSMENT | MANOEUVRE_REVIEW",
"data_freshness_hours": null,
"evidence_completeness": null,
"forecast_stability": "STABLE | SENSITIVE | NOT_ESTIMABLE",
"cutoff_utc": "ISO-8601",
"eligible_message_count": 0,
"dataset_version": "version",
"model_version": "version",
"prediction_timestamp_utc": "ISO-8601"
}
```

These are interface placeholders, not real predictions or selected thresholds.

Appendix B. Prompt for another AI coding assistant

Attach this PDF and give the following prompt to the coding model. Require it to inspect actual sample data before choosing field names or claiming model performance.

You are the senior engineer helping a three-person beginner team build ORBIT PULSE, the SW-5 Satellite Conjunction Risk Intelligence & Decision Support prototype. Treat the attached ORBIT PULSE Final Prototype Master Blueprint as the source of truth. Follow its scope, architecture, UI specification, cutoff rules, evaluation plan and safety language.

Start by inspecting the actual dataset/sample CDM files and confirming event identifier, approved message-availability timestamp, TCA, final target definition, units, schema, licence and permission. Do not assume fields or invent data. If no real CDM dataset is available, create a clearly labelled synthetic fixture for UI and pipeline testing only; do not report synthetic predictive performance as real validation.

Implement in order: (1) CSV/sample ingestion and validation; (2) event grouping, chronological ordering and strict TCA–48h snapshot builder; (3) automated boundary, grouping, duplicate and leakage tests; (4) last-eligible-Pc baseline and one suitable scikit-learn model; (5) chronological event-separated evaluation with false-negative/false-positive reporting; (6) polished Streamlit + Plotly dashboard; (7) honest LIVE / REPLAY / MANUAL

modes with live-updating charts; (8) separate freshness, completeness and forecast-stability indicators; (9) validation-selected advisory decision policy; (10) audit metadata and reproducible artefacts.

Do not start with React, complex APIs, 3D orbital graphics or manoeuvre simulation. Complete the end-to-end MVP first. Do not call final reported CDM Pc an actual collision outcome. Do not call model sensitivity physical orbit uncertainty. Do not compute Pc from miss distance alone. Do not present autonomous burn commands. Explain files, commands and implementation steps for beginners. Produce runnable code, tests, requirements.txt, README and a clean demo flow. Ask one concise question at a time when a missing fact blocks implementation.

Source integration note

Merged from the user-provided ORBIT_PULSE research/technical blueprint (concept, differentiation, evidence intelligence, scenario and debris-protection extensions) and SW-5 final prototype documentation (implementation sequence, ML pipeline, dashboard views, stack, testing, acceptance criteria and demo script). Details requiring official competition rules or real data confirmation are explicitly marked as assumptions.